LAB 2

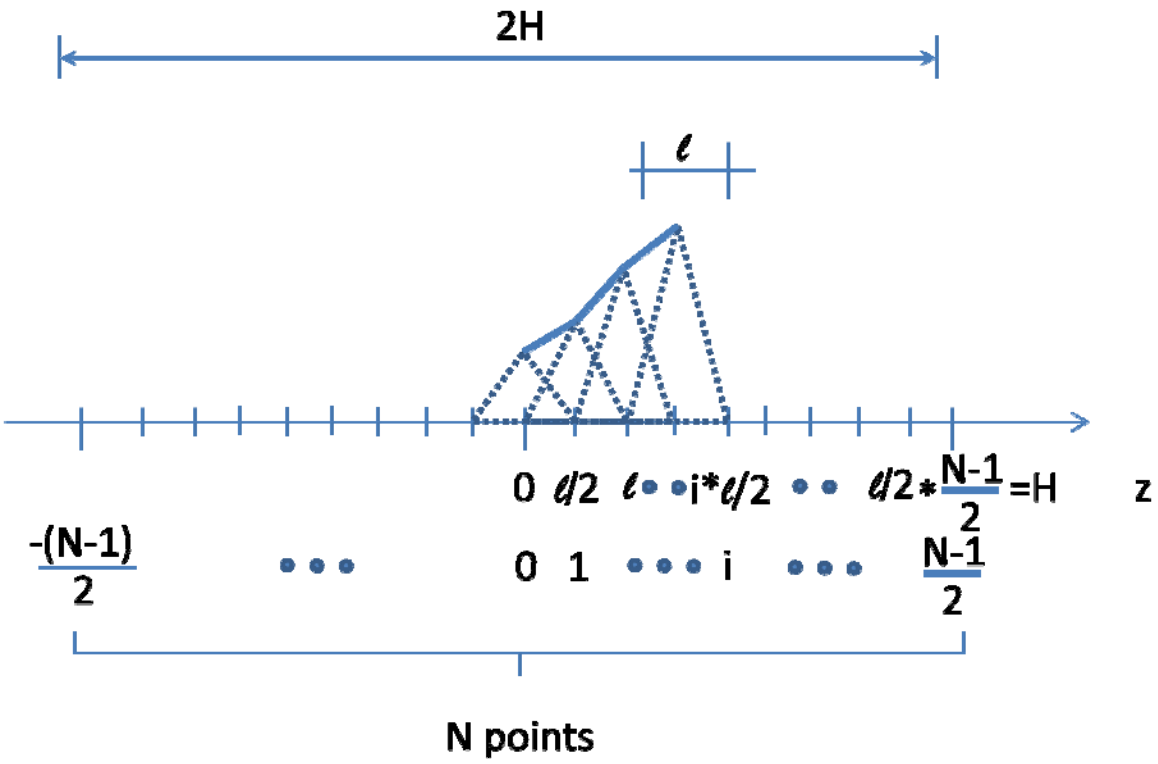
Numerical Techniques: Method of Moments.

Objective

In this lab session you will write a Matlab script to obtain the current distribution on a wire antenna using the Method of Moments.

Pre-Work (to be done before coming to the lab)

You will solve Pocklington's integral equation for the electric field. This is special case of the Electric Field Integral Equation (EFIE) for wire antennas. The basic procedure has been explained in class, but for a good implementation of the algorithm you have to understand well the geometry and how it is discretized.



According to the figure:

- You will analyze a wire (dipole) of length 2H fed in the center.
- You will discretize the geometry in N points that define N-1 segments of length $\ell/2$.
- The basis functions are triangle functions with a base of length ℓ.

Answer the following questions:

- The number of points N should be an odd or an even number?
- Express ℓ as function of N and H.
- Express the coordinate z of a sample point as a function of N, H and the index i (notice that in matlab the index i € [1,N]).

Lab work

You will have to write a Matlab script that analyzes dipole antennas using the method of moments. The input parameters are H (in wavelength), the radius a (in wavelength) and N the number of points. The output is a vector of length N with the current distribution and the antenna input impedance.

The function z12_dip computes the mutal impedance between two current segments with triangular current distribution as a function of the distance.

1) Plot the real and imaginary part of the mutual impedance of two segments of length $\lambda/10$ as function of the distance between their centers from 0 to 3 λ.
2) To write the script follow these steps
   - Define the input parameters, N, H, a.
   - Define a square matrix Z of dimension N full of zeros.
   - Define a column matrix V of length N full of zeros.
   - Compute the value of ℓ
   - Set the center value of V to 1.
   - Create a vector of N elements containing the z position (in wavelength) of each sample point.
   - Fill up the matrix Z using the function Z12_dip.
   - Solve the linear system.
   - Plot the current distribution.
   - Find the input impedance.
3) Check your results for a half-wave dipole.
4) Compare the current distribution for a dipole H=0.1 λ, 0.25 λ and 5/8 λ with the transmission line model.
5) Use your scripts from the previous lab to obtain the radiation pattern and directivity.
6) Obtain a plot for the directivity as function of $2H/\lambda$ from 0.1 to 1.5 using the transmission line model and the MoM.